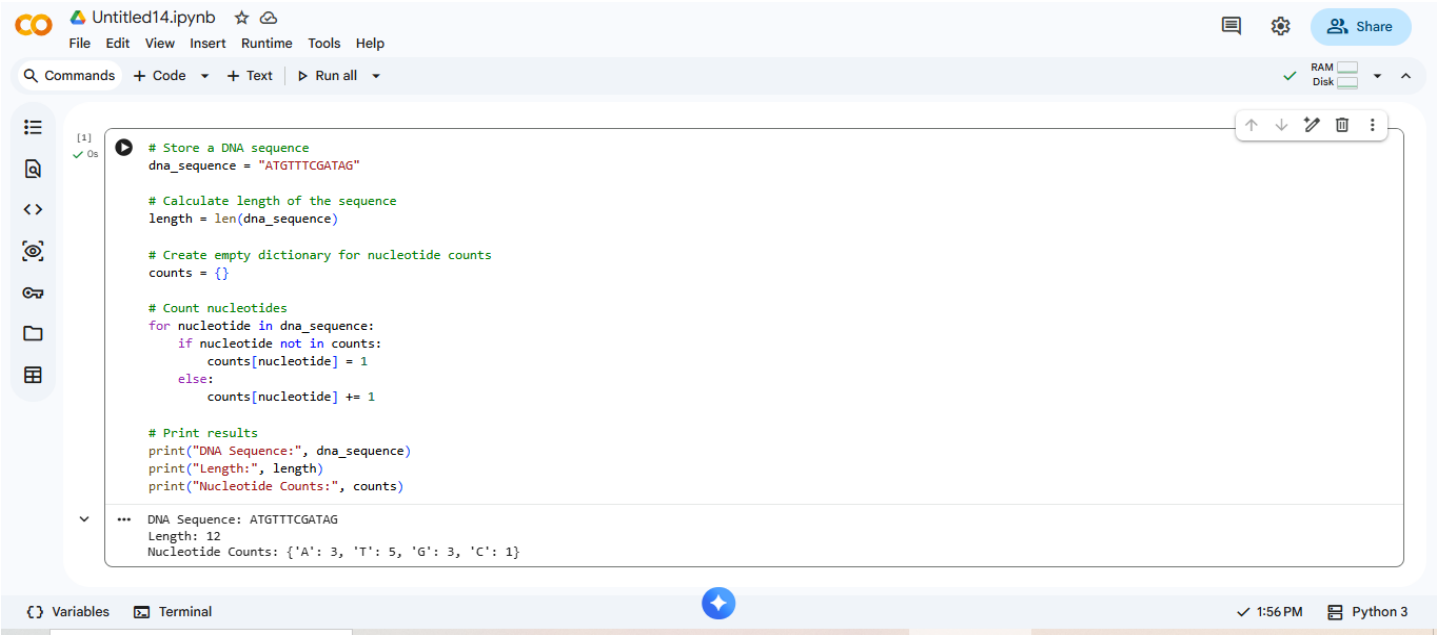
Python Track Capstone Project - Assignment 1

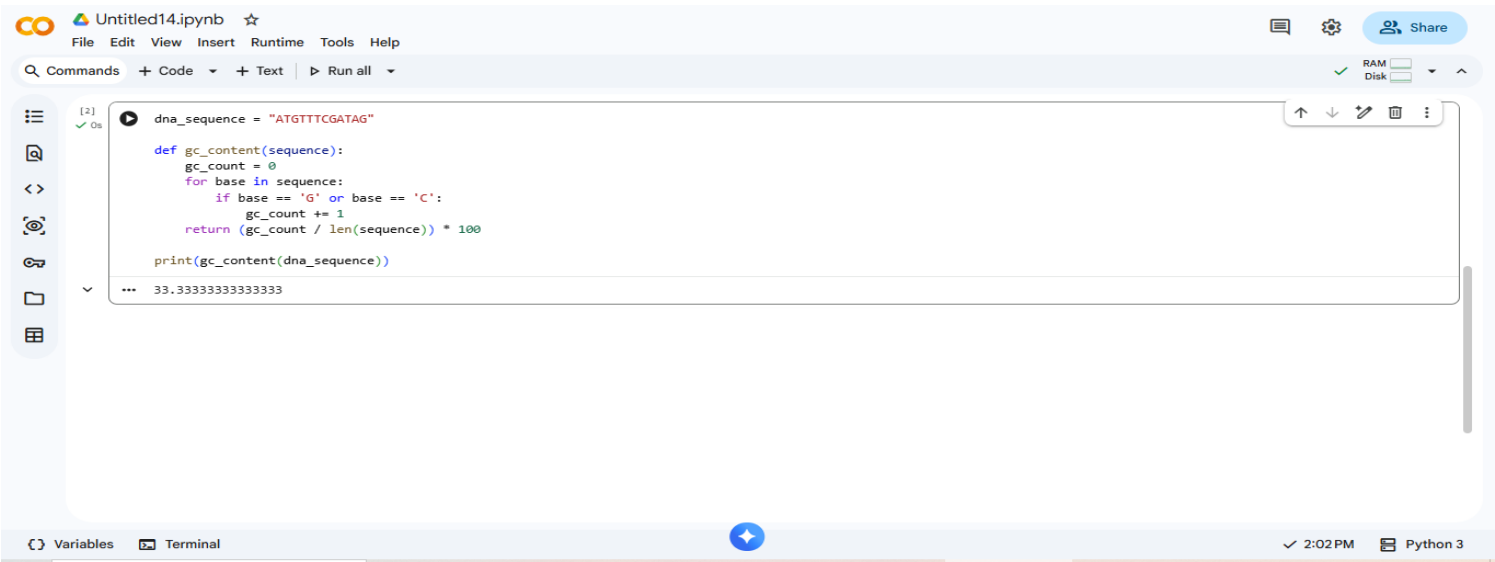
Task 1: Variables and Data Structures

```
# Store a DNA sequence
dna_sequence = "ATGTTTCGATAG"

# Calculate length of the sequence
length = len(dna_sequence)

# Create empty dictionary for nucleotide counts
counts = {}

# Count nucleotides
for nucleotide in dna_sequence:
    if nucleotide not in counts:
        counts[nucleotide] = 1
    else:
        counts[nucleotide] += 1

# Print results
print("DNA Sequence:", dna_sequence)
print("Length:", length)
print("Nucleotide Counts:", counts)
```



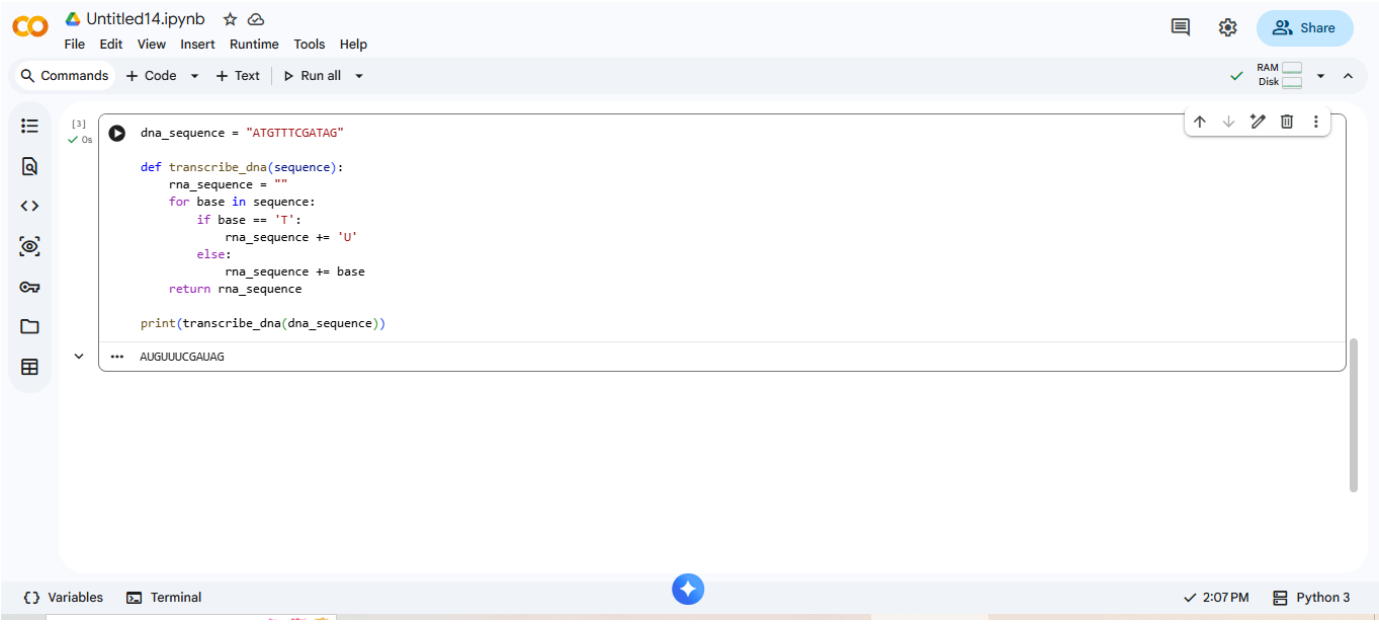
Task 2: GC Content Function

```
dna_sequence = "ATGTTTCGATAG"

def gc_content(sequence):
    gc_count = 0
    for base in sequence:
        if base == 'G' or base == 'C':
            gc_count += 1
    return (gc_count / len(sequence)) * 100

print(gc_content(dna_sequence))
```



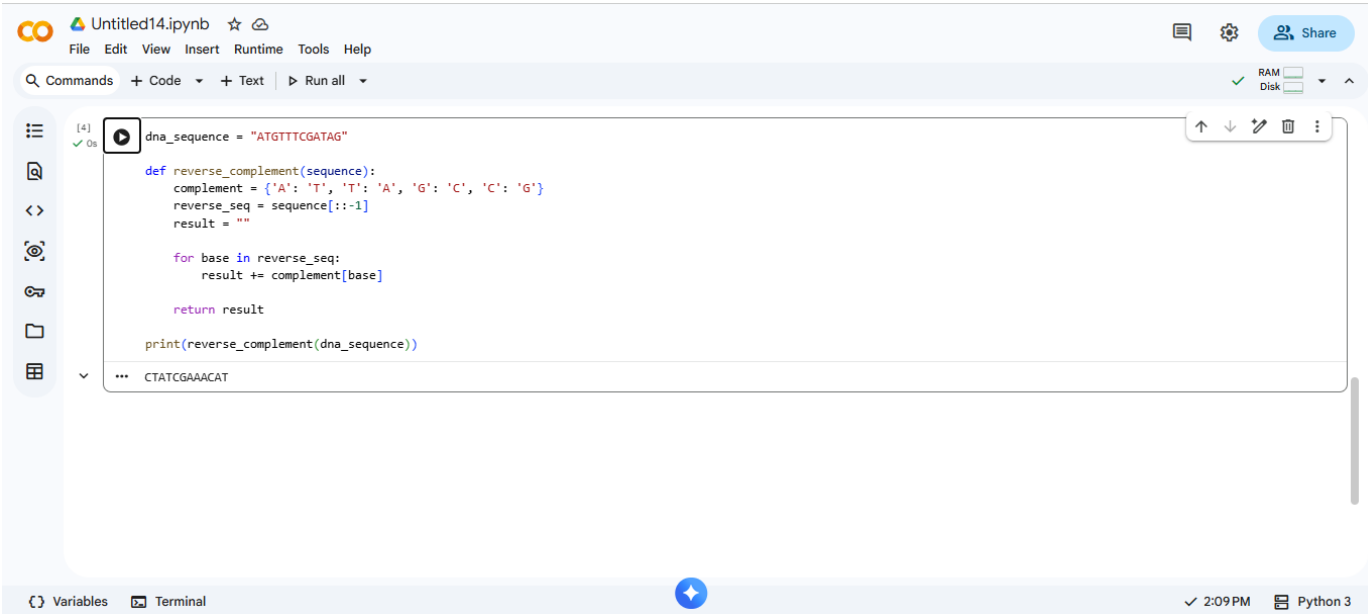
Task 3: DNA Transcription

```
dna_sequence = "ATGTTTCGATAG"

def transcribe_dna(sequence):
    rna_sequence = ""
    for base in sequence:
        if base == 'T':
            rna_sequence += 'U'
        else:
            rna_sequence += base
    return rna_sequence

print(transcribe_dna(dna_sequence))
```



Task 4: Reverse Complement

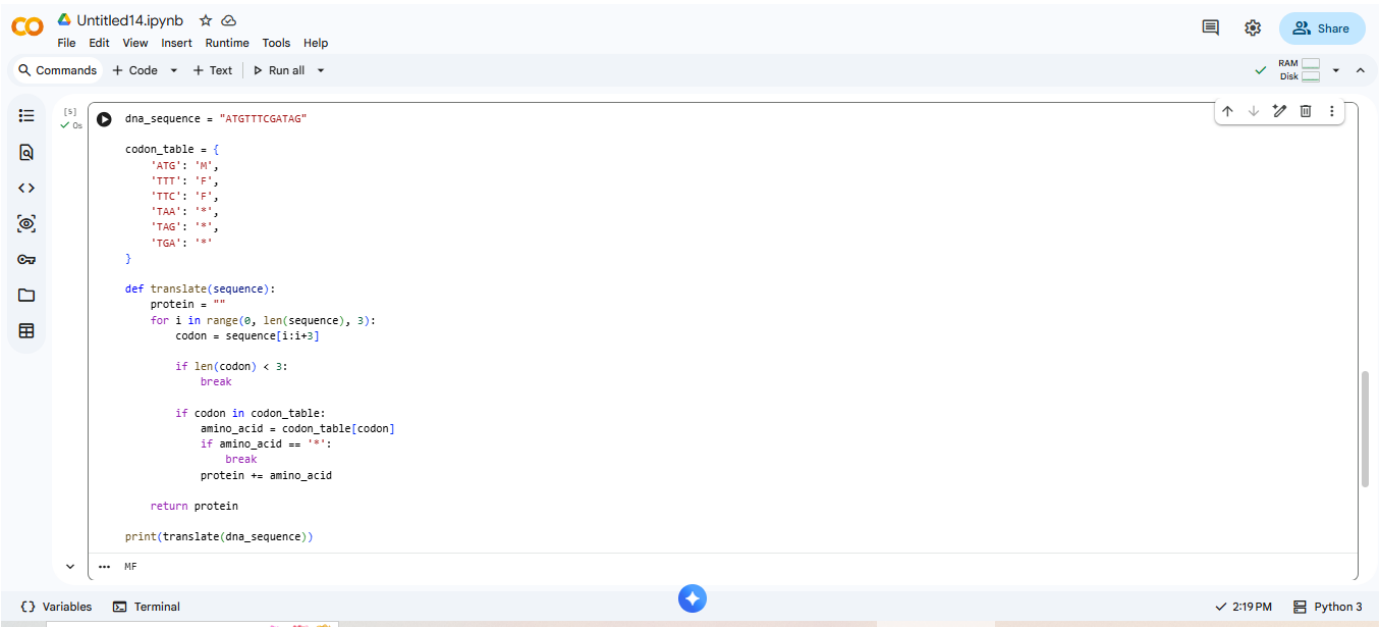
```
dna_sequence = "ATGTTTCGATAG"

def reverse_complement(sequence):
    complement = {'A': 'T', 'T': 'A', 'G': 'C', 'C': 'G'}
    reverse_seq = sequence[::-1]
    result = ""

    for base in reverse_seq:
        result += complement[base]

    return result

print(reverse_complement(dna_sequence))
```



Task 5: DNA Translation

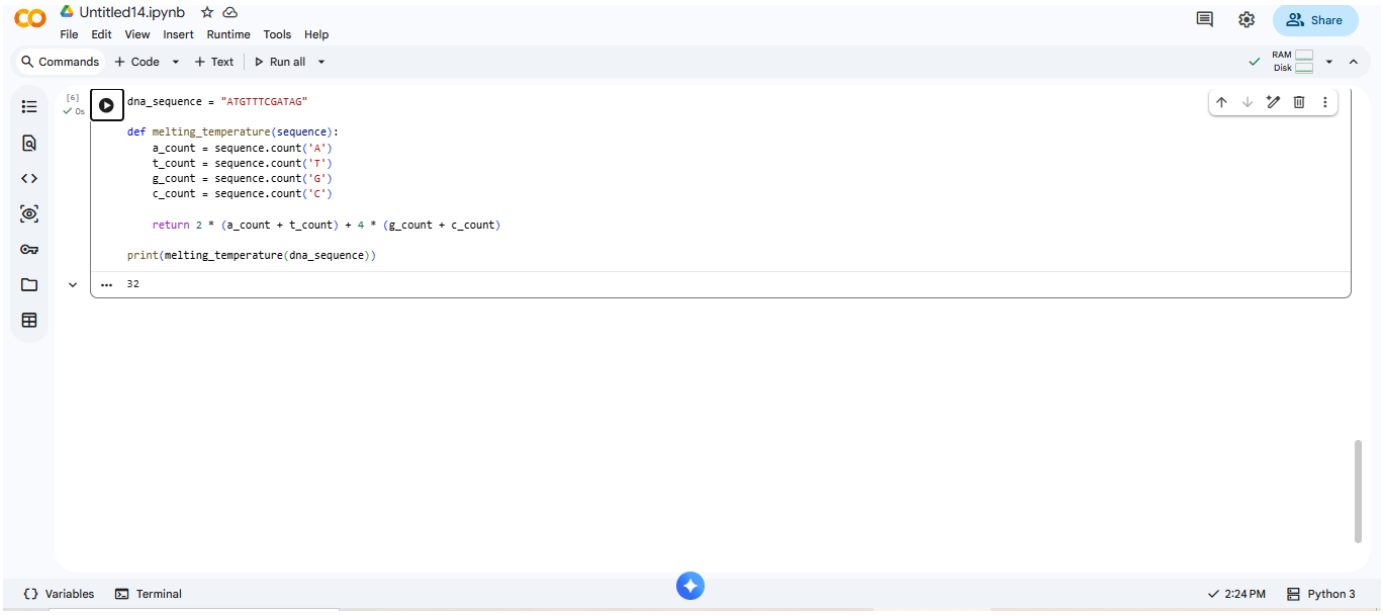
```
dna_sequence = "ATGTTTCGATAG"

codon_table = {
    'ATG': 'M',
    'TTT': 'F',
    'TTC': 'F',
    'TAA': '*',
    'TAG': '*',
    'TGA': '*'
}

def translate(sequence):
    protein = ""
    for i in range(0, len(sequence), 3):
        codon = sequence[i:i+3]

        if len(codon) < 3:
            break

        if codon in codon_table:
            amino_acid = codon_table[codon]
            if amino_acid == '*':
                break
```

```
            protein += amino_acid

        return protein

    print(translate(dna_sequence))
```



Task 6: Melting Temperature (Tm) Calculation

```
dna_sequence = "ATGTTTCGATAG"

def melting_temperature(sequence):
    a_count = sequence.count('A')
    t_count = sequence.count('T')
    g_count = sequence.count('G')
    c_count = sequence.count('C')

    return 2 * (a_count + t_count) + 4 * (g_count + c_count)

print(melting_temperature(dna_sequence))
```



Task 7: Conditional Statements and Validation

```
dna_sequence = "ATGTTTCGATAG"

def is_valid_dna(sequence):
    for base in sequence:
        if base != 'A' and base != 'T' and base != 'G' and base != 'C':
            return False
    return True

print(is_valid_dna(dna_sequence))
```

Final Task: Integrative Analysis

```
dna_sequence = "ATGTTTCGATAG"

if is_valid_dna(dna_sequence):
    print("GC Content:", gc_content(dna_sequence))
    print("RNA Sequence:", transcribe_dna(dna_sequence))
    print("Reverse Complement:", reverse_complement(dna_sequence))
    print("Protein Sequence:", translate(dna_sequence))
    print("Melting Temperature:", melting_temperature(dna_sequence))
else:
    print("Invalid DNA sequence")
```

Untitled14.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Share

RAM Disk

[9]

```python
dna_sequence = "ATGTTTCGATAG"

if is_valid_dna(dna_sequence):
    print("GC Content:", gc_content(dna_sequence))
    print("RNA Sequence:", transcribe_dna(dna_sequence))
    print("Reverse Complement:", reverse_complement(dna_sequence))
    print("Protein Sequence:", translate(dna_sequence))
    print("Melting Temperature:", melting_temperature(dna_sequence))
else:
    print("Invalid DNA sequence")
```

```
GC Content: 33.33333333333333
RNA Sequence: AUGUUUCGAUAG
Reverse Complement: CTATCGAAACAT
Protein Sequence: MF
Melting Temperature: 32
```

Variables Terminal

2:36 PM Python 3